MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION (2024 ADMISSION ONWARDS)

MG5DSCMAT302 – FUNDAMENTALS OF GROUPS AND RINGS

MODEL QUESTION PAPER

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Answer any **5** questions. Each question carries **2** marks.

1. Let . be the usual multiplication operation on $\mathbb{Z}$ the set of integers, and let $H = \{n^2 | n \in \mathbb{Z}^+\}$. Determine whether H is closed under multiplication. [U][1]

2. Find the identity element of the set $\mathbb{Z}$ with binary operation $*$ defined by $a * b = ab + a + b$. [U][1]

3. List the generators of (a) $\mathbb{Z}$ (b) $\mathbb{Z}_5$ [U][2]

4. Find the quotient and remainder, according to the division algorithm, when -42 is divided by 9. [U][2]

5. List the elements of the subgroup generated by $\{12, 30\}$ of $\mathbb{Z}_{36}$. [U][3]

6. State the theorem of Lagrange. [K][3]

7. Define ring homomorphism. [K][4]

8. Find the characteristic of the ring $\mathbb{Z}_6$ x $\mathbb{Z}_{15}$. [K][4]

[2 × 5 = 10]

Part B

Answer any **5** questions. Each question carries **6** marks.

9. Show that Q^+ with the operation $*$ defined by $a * b = ab/2$ is a group. [U][1]

10. Show that $< n\mathbb{Z}, + >$ is isomorphic to $< \mathbb{Z}, + >$. [U][1]

11. Compute σ^{100} given $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$. [A][5]

12. Give the subgroup diagram of the subgroups of $\mathbb{Z}_{12}$. [U][2]

13. Find all orbits of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 3 & 6 & 2 & 4 \end{pmatrix}$. [A][3]

14. Find all cosets of the subgroup $4\mathbb{Z}$ of $\mathbb{Z}$. [U][3]

15. Determine whether $\mathbb{Z}$ x $\mathbb{Z}$ is a ring with addition and multiplication by components. [U][4]

16. Prove every finite integral domain is a field. [U][4]

[6 × 5 = 30]

Part C

Answer any **3** questions. Each question carries **10** marks.

17. (a) Show that the left and right cancellation laws hold in a group G with binary operation $*$.

    (b) Prove that the identity element and inverse of each element are unique in a group.

    [U][1]

18. (a) Prove a subgroup of a cyclic group is cyclic.

    (b) Describe all cyclic groups upto isomorphism.

    [U][2]

19. (a) Let G be a cyclic group with n elements and generated by a. Let $b \in G$ and let $b = a^s$. Prove that b generates a cyclic subgroup H of G containing n/d elements, where d is the greatest common divisor of n and s.

    (b) Find the order of all the subgroups of $\mathbb{Z}_{18}$.

    [U][2]

20. Let ϕ be a homomorphism of a group G into a group G'. Prove

    (a) If e is the identity element in G, then $\phi(e)$ is the identity element e' in G'.

    (b) If $a \in G$, then $\phi(a^{-1}) = \phi(a)^{-1}$.

    (c) If H is a subgroup of G, then $\phi(H)$ is a subgroup of G'.

(d) If K' is a subgroup of G', then $\phi^{-1}(K')$ is a subgroup of G.

[U][3]

21. Let R be a ring with additive identity 0, and let $a, b \in R$. Prove

(a) $0a = a0 = 0$

(b) $a(-b) = (-a)b = -(ab)$

(c) $(-a)(-b) = ab$.

[U][4]

22. (a) Solve the equation $x^2 - 5x + 6 = 0$ in $\mathbb{Z}_{12}$.

(b) Classify each nonzero element of $\mathbb{Z}_{20}$ as a unit or a 0 divisor.

(c) Prove the cancellation laws hold in a ring R if and only if R has no divisors of 0.

[A][4]

[10 × 3 = 30]